

Machine Learning Notes

Fall 2025

Your Name

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Chapter 1

Introduction

The Real Explanation

Formally, let $\mathcal{D} = \{(x_i, y_i)\}_{i=1}^n$ be a dataset. We seek $f : \mathcal{X} \rightarrow \mathcal{Y}$ that minimizes $\mathbb{E}_{(x,y) \sim \mathcal{D}}[\ell(f(x), y)]$.

1.1 Key Concepts

Key Takeaways

- Supervised vs. unsupervised learning
- Bias-variance tradeoff
- Overfitting and regularization

Key Takeaways

this is test